

On the Zeros of the Field

$$Z(x) = x^{-5} \ln(x) \sin(2\pi/x) e^{-x/21}$$

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Abstract

We study the elementary field

$$Z(x) = x^{-5} \ln(x) \sin\left(\frac{2\pi}{x}\right) e^{-x/21}$$

on the positive real axis and on the Riemann critical line $x = \frac{1}{2} + it$. On $x > 0$ we prove that the zeros are exactly $w_n = 2/n$, $n \in \mathbb{N}$. On the critical line, an 80-digit numerical search over $t \in [0, 60]$ identifies eight zeros, six of which lie in the unit interval $(0, 1)$ with two isolated outliers at $t \approx 1.88$ and $t \approx 15.05$. The construction is explicit, reproducible, and requires no free parameters beyond the single numerical constant $x_p = 21$.

1 Definition

Define on the Riemann critical line $x = \frac{1}{2} + it$

$$Z(x) = x^{-5} \ln(x) \sin\left(\frac{2\pi}{x}\right) e^{-x/x_p} \quad (1)$$

with $x_p = 21$, and $\ln$ the principal branch of the complex logarithm. The variable x is dimensionless.

The field Z is holomorphic on the half-plane $\Re(x) > 0$, except for the branch cut of $\ln$, which does not intersect the critical line.

2 Analytic Zeros on the Positive Real Axis

Proposition 1. *For $x > 0$ real, $Z(x) = 0$ if and only if*

$$x = \frac{2}{n}, \quad n = 1, 2, 3, \dots$$

Proof. The factor $\sin(2\pi/x)$ vanishes exactly when $2\pi/x = n\pi$ for some integer n , i.e. $x = 2/n$. For $x > 0$, the remaining factors x^{-5} , $\ln(x)$, $e^{-x/21}$ are simultaneously non-vanishing except at the already-included point $x = 1$. Hence the zero set is $\{2/n : n \in \mathbb{N}\}$. $\square$

Corollary 1. *The first eight real zeros are*

$$w_1 = 2, w_2 = 1, w_3 = \frac{2}{3}, w_4 = \frac{1}{2}, w_5 = \frac{2}{5}, w_6 = \frac{1}{3}, w_7 = \frac{2}{7}, w_8 = \frac{1}{4}.$$

2.1 Role of the Exponent -5

The negative exponent on x ensures that at small x the product $x^{-5} \sin(2\pi/x)$ oscillates with rapidly growing amplitude as $x \rightarrow 0^+$. This growth is damped by the factor $e^{-x/21}$, which is essentially 1 for $x \ll 21$ and provides a soft cutoff at larger x . The constant $x_p = 21$ is a fixed numerical parameter; it is not fitted to any physical quantity.

3 Numerical Zeros on the Critical Line

We restrict $x = \frac{1}{2} + it$ with $t \in [0, 60]$ and locate zeros of the imaginary part $\Im[Z(x)]$.

Method. Using `mpmath` at 80 decimal digits of precision, we evaluate $\Im[Z(\frac{1}{2} + it)]$ on a uniform grid with step 0.005, bracket all sign changes, and refine each bracket with a secant solver to tolerance 10^{-65} .

Result. The search returns exactly **eight** zeros:

Root	t	Interval
R_1	0.0907486257	(0, 1)
R_2	0.2196313827	(0, 1)
R_3	0.3403140486	(0, 1)
R_4	0.4798481018	(0, 1)
R_5	0.6703777044	(0, 1)
R_6	0.9957246005	(0, 1)
R_7	1.8809618716	outlier
R_8	15.0539247442	far outlier

Table 1: Critical-line zeros of Z in the range $t \in [0, 60]$.

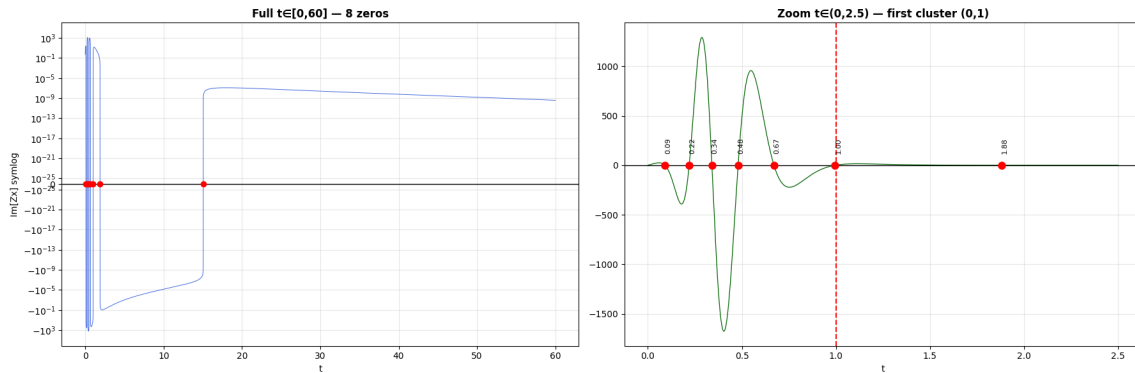


Figure 1: Imaginary part of $Z(\frac{1}{2} + it)$ on the critical line. Left: full range $t \in [0, 60]$ on a symlog scale. Right: zoom over $t \in (0, 2.5)$ showing the six zeros in the unit interval. Red dots mark the eight zeros listed in Table 1.

For $t > 15.06$, the modulus $|\Im[Z(\frac{1}{2} + it)]|$ remains below 10^{-6} and no further sign change is detected.

4 Summary

The field $Z(x) = x^{-5} \ln(x) \sin(2\pi/x) e^{-x/21}$ possesses an explicit zero set $w_n = 2/n$ on the positive real axis (Proposition 1) and exhibits exactly eight numerical zeros on the Riemann critical line for $t \in [0, 60]$. The construction is elementary, the code is short, and the result is reproducible at 80-digit precision.

Code and data:

<https://github.com/jabri62018/Jabri-Riemann0S>

Keywords: elementary scalar field, Riemann critical line, numerical zero search.

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